

STUDENT NUMBER: _____



Pymble Ladies' College

2022 Chemistry Trial Examination

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA approved calculators may be used
- A formulae sheet, data sheet and Periodic Table are provided separately
- For questions in Section I, record your responses on the multiple choice answer sheet provided at the back of this paper.
- For questions in Section II, show all relevant working in questions involving calculations.

Total Marks: 100

Section I – 20 marks

- Attempt Questions 1 – 20
- Allow about 35 minutes for this section

Section II – 80 marks

- Attempt Questions 21 – 34
- Allow about 2 hours and 25 minutes for this section

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Section I

20 marks

Attempt Questions 1 – 20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 20.

1. Which of the following substances would make the best primary standard?

	<i>Substance</i>	<i>Purity</i>	<i>Solubility</i>	<i>Formula mass</i>
A.	P	high	low	low
B.	Q	high	high	high
C.	R	high	low	high
D.	S	high	high	low

2. An acid is a substance that donates one or more protons”
Who first gave this definition of an acid?

- A. Arrhenius
- B. Brønsted-Lowry
- C. Lewis
- D. Lavoisier

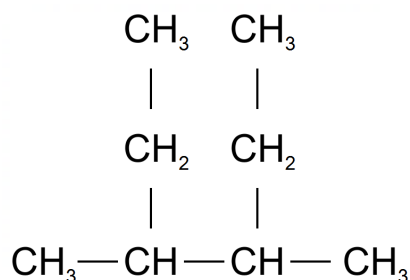
3. The table below shows the pH range of four acid-base indicators

Indicator	pH range
W	3.3 - 4.3
X	6.2- 7.5
Y	7.8 -10.3
Z	8.1 - 9.8

Which indicator would be the best choice for a titration between potassium hydroxide and ethanoic acid?

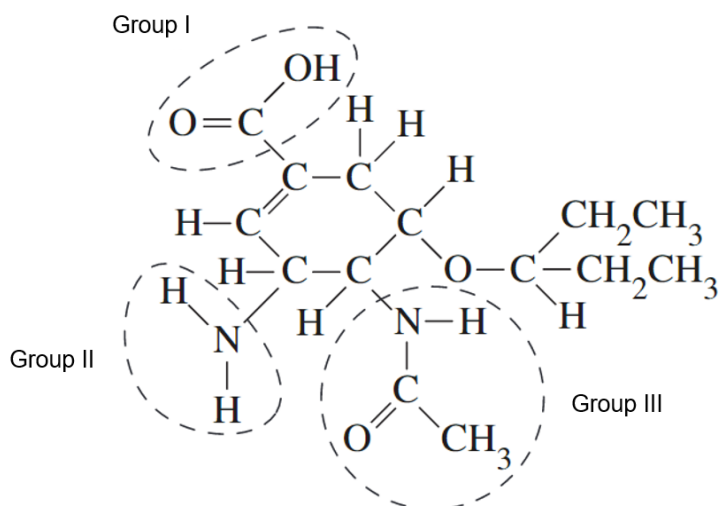
- A. W
- B. X
- C. Y
- D. Z

4. What is the correct systematic name for the following compound?



- A. 2-ethyl-3-methylpentane
- B. 3-methyl-4-ethylpentane
- C. 3,4-dimethylhexane
- D. 2,3-diethylbutane

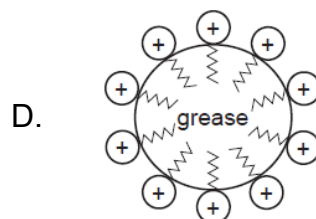
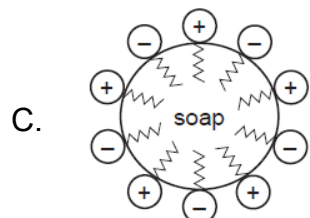
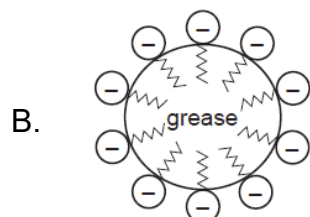
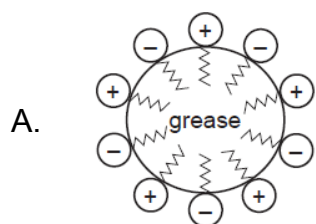
5. The structure of a compound is shown. It has been suggested that this molecule may be an effective drug for treating influenza.



Which row of the table correctly names the functional groups labelled I, II and III?

	Group I	Group II	Group III
A.	aldehyde	amide	ester
B.	carboxylic acid	amine	amide
C.	ketone	amine	carboxylic acid
D.	carboxylic acid	amide	amine

6. Which of the following diagrams represents a micelle that forms in water when soap is used to remove grease from dirty dishes?

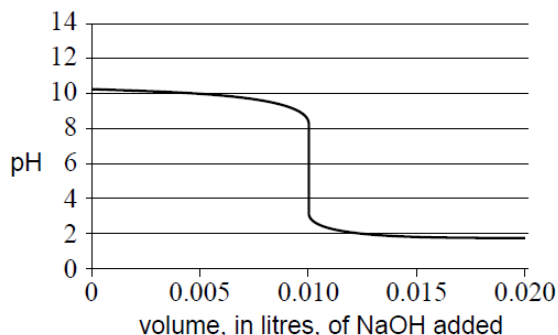


7. How many structural isomers are there for the compound with molecular formula C_3H_6BrCl ?

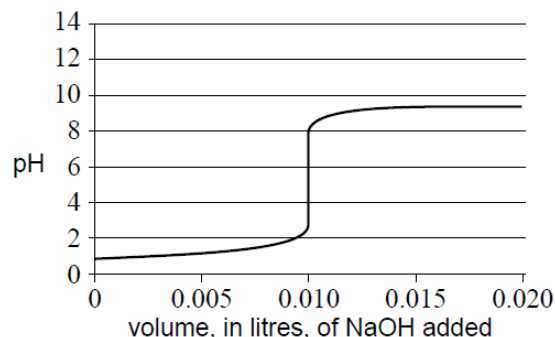
- A. 4
B. 5
C. 6
D. 7

8. Which titration curve best represents the change in pH as 0.100 M NaOH solution is added to a 10.0 mL aliquot of 0.100 M HCl solution?

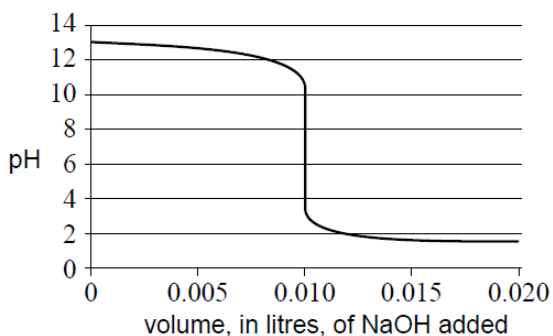
A.



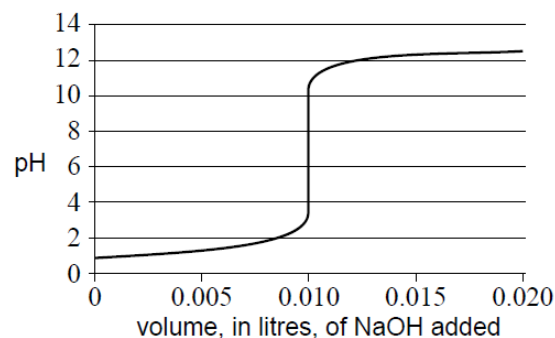
B.



C.



D.



9. A 0.10 M solution of fluoride ions is gradually added to a solution containing Ba^{2+} , Ca^{2+} and Pb^{2+} ions, each at a concentration of 1×10^{-3} M. In what order, from first to last, will the precipitates of BaF_2 , CaF_2 and PbF_2 form?

Solubility Product, K_{sp}	
BaF_2	1.8×10^{-7}
CaF_2	1.5×10^{-10}
PbF_2	7.1×10^{-7}

- A. CaF_2 , PbF_2 , BaF_2
 B. BaF_2 , CaF_2 , PbF_2
 C. PbF_2 , BaF_2 , CaF_2
 D. CaF_2 , BaF_2 , PbF_2

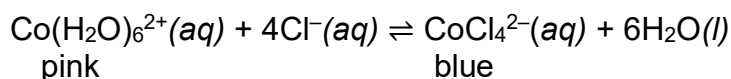
10. Methanoic acid and ethanoic acid are both weak acids with the following acid dissociation constants, K_a , at 25°C:

- methanoic acid (HCOOH) = 1.82×10^{-4}
- ethanoic acid (CH_3COOH) = 1.74×10^{-5}

Two separate solutions were prepared, one of 0.1 M methanoic acid and the other of 0.1 M ethanoic acid. Which one of the following would be present in the lowest concentration at 25°C?

- A. CH_3COOH in the ethanoic acid solution
- B. CH_3COO^- in the ethanoic acid solution
- C. HCOOH in the methanoic acid solution
- D. HCOO^- in the methanoic acid solution

11. A solution contains an equilibrium mixture of two different cobalt (II) ions.



The solution contains pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ ions and blue CoCl_4^{2-} ions, and the solution has a purple colour.

10 mL of the purple solution was poured into a test tube.

Which one of the following changes would cause 10 mL of the purple cobalt (II) ion solution to turn blue?

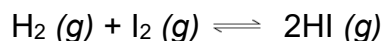
- A. the addition of a few drops of 10 M hydrochloric acid at a constant temperature
- B. the addition of a few drops of 0.1 M silver nitrate at a constant temperature
- C. the addition of a few drops of a catalyst at a constant temperature
- D. the addition of a few drops of water at a constant temperature

12. One litre of an aqueous solution of potassium hydroxide (KOH) has a pH of 12.0 at 25°C.

How many moles of pure HCl gas must be added to the solution to lower the pH from pH 12.0 to 2.0?

- A. 10
- B. 2
- C. 0.02
- D. 0.01

13. Hydrogen iodide is produced by the reaction between hydrogen and iodine.

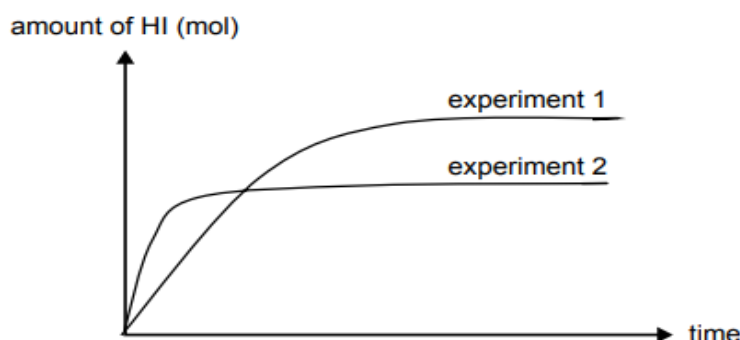


Two experiments were conducted.

Experiment 1: quantities of $\text{H}_2(g)$ and $\text{I}_2(g)$ were placed in a sealed vessel and the reaction allowed to proceed at constant temperature.

Experiment 2: experiment 1 was repeated, but at a different temperature.

The graph below shows the amount of hydrogen iodide produced over the course of experiments 1 and 2.



These results show that experiment 2 was conducted at a

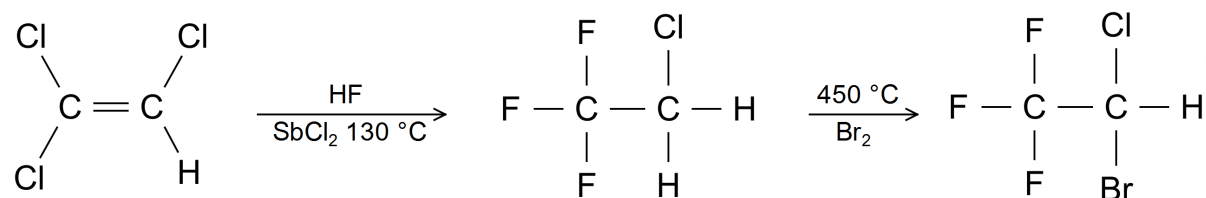
- A. lower temperature than experiment 1 and the reaction is endothermic.
- B. lower temperature than experiment 1 and the reaction is exothermic.
- C. higher temperature than experiment 1 and the reaction is endothermic.
- D. higher temperature than experiment 1 and the reaction is exothermic.

14. Which one of the following statements about 10.0 mL of 0.10 M HCl and 10.0 mL of 0.10 M CH_3COOH solutions is true?

- A. Each solution will react completely with 10.0 mL of 0.10 M NaOH solution.
- B. Each solution will have the same electrical conductivity.
- C. Each solution will react at the same rate with 1.00 g of magnesium ribbon.
- D. The concentration of H_3O^+ ions will be greater in the CH_3COOH solution.

15. Haloethane (1-bromo-1-chloro-2,2,2-trifluoroethane) is a general anaesthetic.

The following diagram represents the reaction pathway that produces haloethane.



molecule X

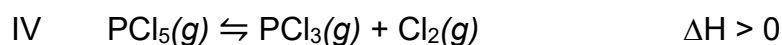
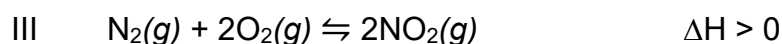
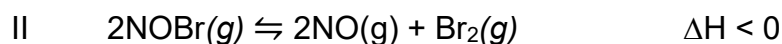
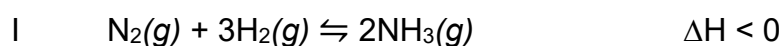
step 1

step 2

Which of the following correctly identifies molecule X and the type of reaction shown in step 2?

	<i>Molecule X</i>	<i>Type of reaction in step 2</i>
A	1,2,2-trichloroethane	Addition
B	1,1,2-trichloroethane	Substitution
C	1,1,2-trichloroethene	Substitution
D	1,1,2-trichloroethene	Addition

16. Four different equilibrium systems (I–IV) at standard pressure and temperature are shown.



In each system, the pressure was increased. Later, the temperature was decreased.

In which equilibrium system would BOTH changes result in a product yield increase?

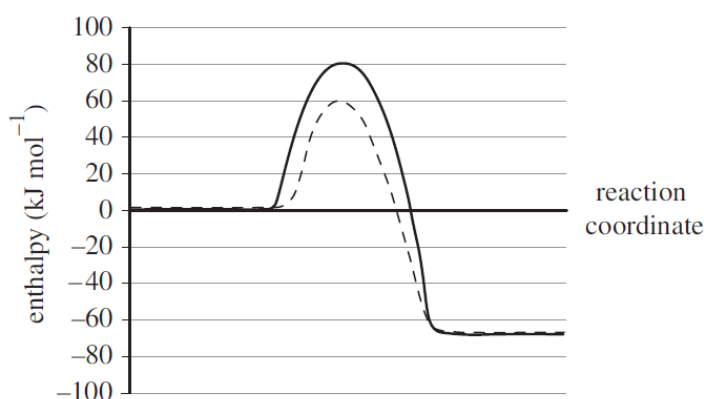
- A. I
- B. II
- C. III
- D. IV

17. A student performed an experiment in which 250 mL of water was heated by burning methanol in a spirit burner. The enthalpy of combustion of methanol is -726 kJ mol^{-1} . Assume that all heat released by the process is absorbed by the water.

What is the maximum change in temperature of the water when 1.26 g of methanol is completely combusted?

- A. $0.434 \text{ }^{\circ}\text{C}$
- B. $3.66 \text{ }^{\circ}\text{C}$
- C. $27.3 \text{ }^{\circ}\text{C}$
- D. $36.3 \text{ }^{\circ}\text{C}$

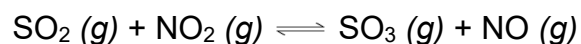
18. A particular reaction can occur with or without a catalyst. The energy profiles of this reaction, both catalysed and uncatalysed, are shown.



Which row of the table best matches the reactions as shown by the energy profiles?

	E_a of uncatalysed reaction (kJ mol^{-1})	ΔH of uncatalysed reaction (kJ mol^{-1})	E_a of catalysed reaction (kJ mol^{-1})	ΔH of catalysed reaction (kJ mol^{-1})
A.	-80	70	-60	70
B.	80	-150	80	60
C.	80	-70	60	-70
D.	20	80	150	-80

19. Consider this reaction, whose $K_{\text{eq}} = 33$



If 0.10 mol each of SO_2 and NO_2 are placed in a 1.0 L container, what is the concentration of SO_3 at equilibrium?

- A. 0.0030 M
- B. 0.015 M
- C. 0.085 M
- D. 0.097 M

20. An experiment is conducted in which 4-aminobutanoic acid, $\text{H}_2\text{N}(\text{CH}_2)_3\text{COOH}$, forms a polymer containing 100 monomer units.

What is the molar mass of this polymer?

- A. $8.5 \times 10^3 \text{ g mol}^{-1}$
- B. $1.0 \times 10^4 \text{ g mol}^{-1}$
- C. $1.0 \times 10^2 \text{ g mol}^{-1}$
- D. 103 g mol^{-1}

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Chemistry

Section II Answer Booklet

80 marks

Attempt Questions 21 – 34

Allow about 2 hours and 25 minutes for this section

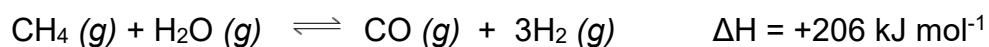
Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Show all relevant working in questions involving calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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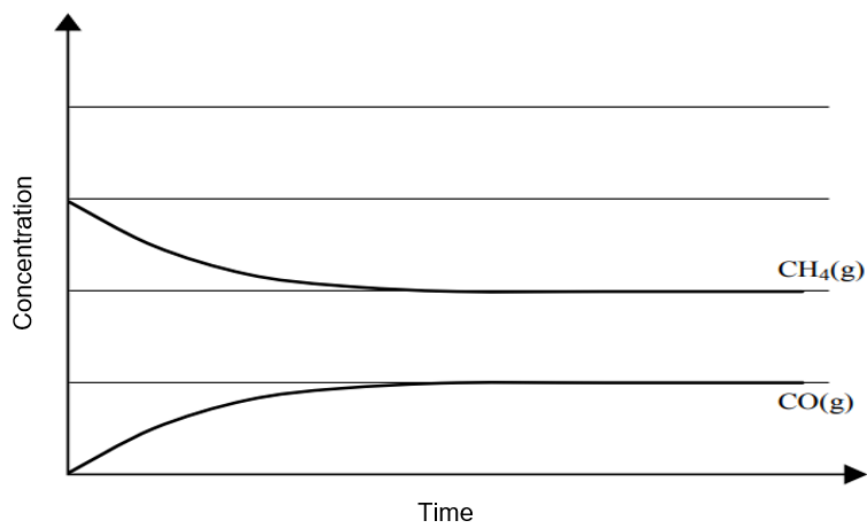
Please turn over

Question 21 (4 marks)

Carbon monoxide and hydrogen can be produced from the reaction of methane with steam, according to the equation:



Some methane and steam were placed in a closed container and allowed to react at a fixed temperature. The following graph shows the changes in concentration of methane and carbon monoxide as the reaction progresses.

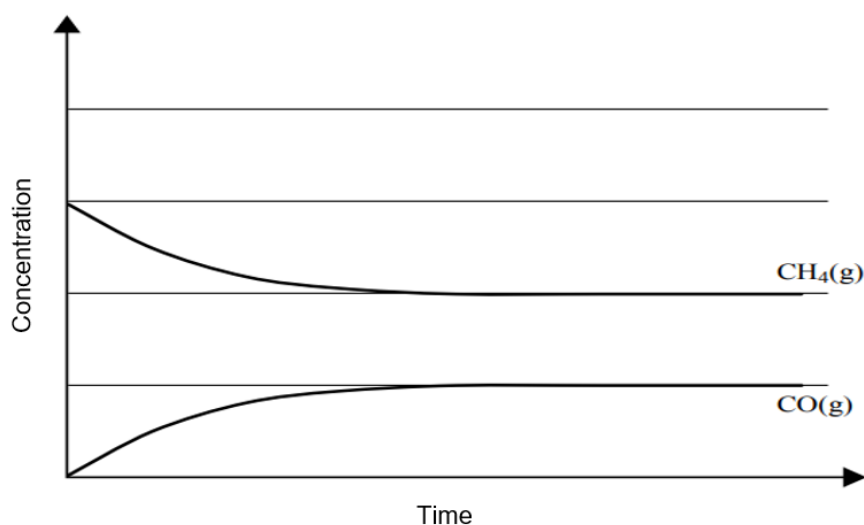


- (a) On the graph above, draw a line to predict the change in concentration of hydrogen gas as the reaction progresses. Label this line as 'H₂'

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- (b) On the graph below, draw a line to show how the formation of carbon monoxide would differ over time in the presence of a catalyst.

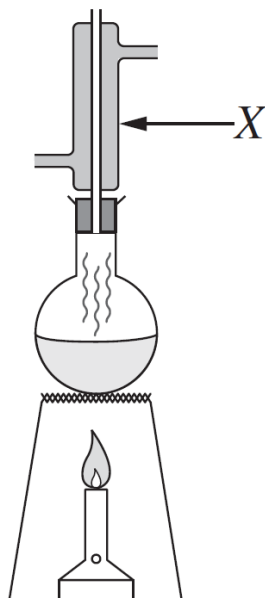
2



Question 22 (4 marks)

A student conducted an experiment in the school laboratory to synthesise an ester.

Some of the equipment used is shown in the diagram below.



- (a) Identify the risk associated with the students experimental set up in the diagram and suggest alternative equipment. **2**

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- (b) Outline how the equipment piece labelled 'X' assisted in maximising the yield of the ester. **1**

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- (c) Identify what else the student could do to maximise the yield of ester. **1**

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Question 23 (5 marks)

Industrially, ethanol, $\text{C}_2\text{H}_5\text{OH}$, is made by either of two methods.

One method uses ethene (ethylene), C_2H_4 , which is derived from crude oil. The other method uses fermentation of a sugar such as glucose, $\text{C}_6\text{H}_{12}\text{O}_6$ or sucrose with yeast in an aqueous solution.

- (a) Write a balanced equation to show the production of $\text{C}_2\text{H}_5\text{OH}$ from $\text{C}_6\text{H}_{12}\text{O}_6$ and yeast. **1**

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- (b) You carried out this fermentation reaction in the laboratory. What conditions were necessary for this fermentation to occur. **2**

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- (c) Determine the mass, in grams, of pure $\text{C}_2\text{H}_5\text{OH}$ that would be produced from 1.25 kg of $\text{C}_6\text{H}_{12}\text{O}_6$ dissolved in water. **2**

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Question 24 (3 marks)

Some salts display amphoteric behaviour.

Using a suitable example, explain what is meant by 'amphoteric'. Support your answer using appropriate equations.

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Question 25 (5 marks)

Polymers are important substances and have many practical applications.

Scientists have developed biopolymers to replace traditional polymers.

One example of this is polyhydroxybutyrate (PHB) that was developed as an alternative to polypropylene (PP)

Production	Produced in the cells of microorganisms such as <i>bacillus sp</i> using agricultural residue such as sugarcane bagasse, a dry pulpy material that remains after crushing sugarcane, as a carbon source.	Fractional distillation of crude oil. Fractions further thermally or catalytically cracked to produce the monomer.
Monomer structure	$ \begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{O} \\ & & & & & & & \\ \text{H} - \text{O} - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} - \text{O} - \text{H} \\ & & & & & & & \\ & \text{H} & & \text{H} - \text{C} - \text{H} & & \text{H} & & \\ & & & & & & & \\ & & & \text{H} & & & & \end{array} $	$ \begin{array}{ccccccc} & \text{H} & & & & & \\ & & & & & & \\ \text{H} - & \text{C} & - & \text{C} = & \text{C} & - & \text{H} \\ & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & \end{array} $
Monomer name	hydroxybutyrate	Propylene

(a) Outline the advantage of using PHB as a replacement for PP.

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(b) What type of polymerisation reaction produces PHB and PP?

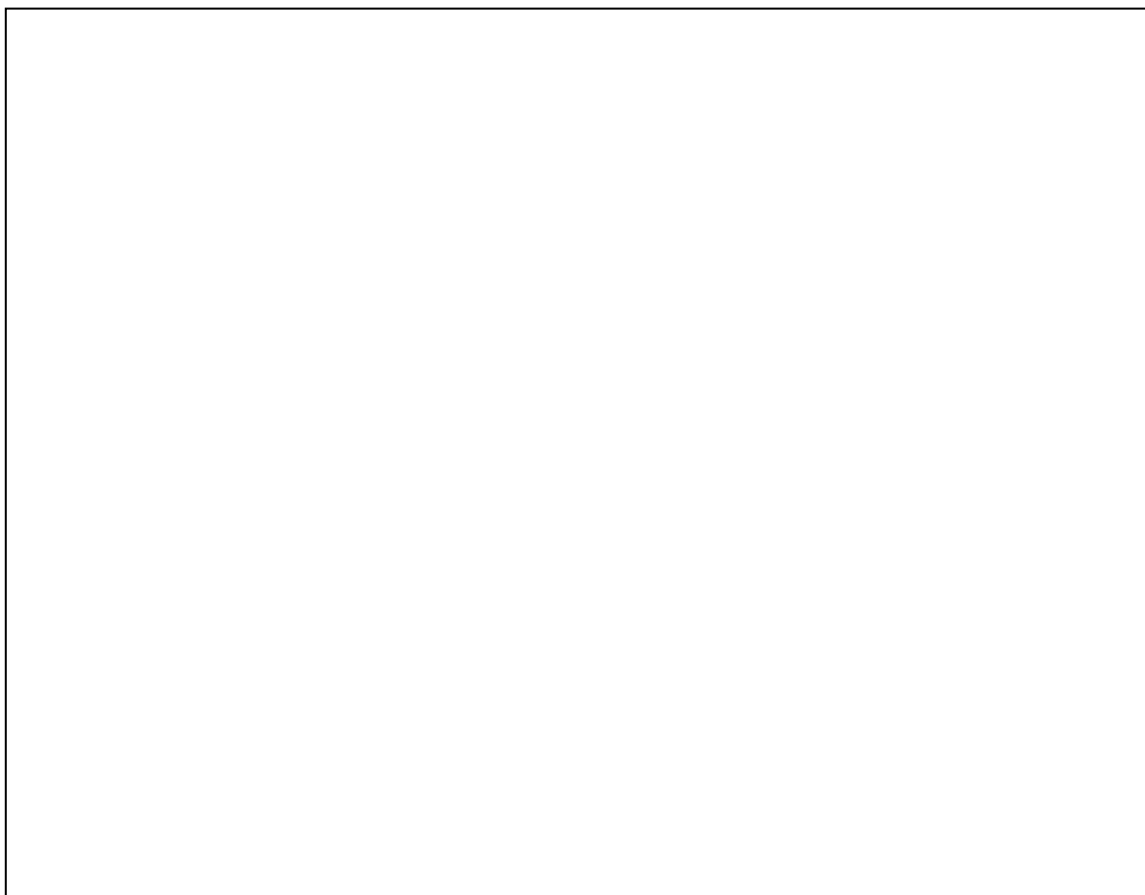
2

Polymer	Type of polymerisation reaction
polyhydroxybutyrate	
polypropylene	

Question 25 continues on page 19

Question 25 (continued)

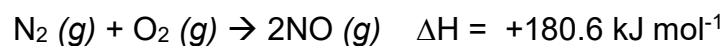
(c) Draw three repeating units for each of the polymers made from the monomers. **2**

A large, empty rectangular box with a thin black border, intended for the student to draw the repeating units of polymers.

End of Question 25

Question 26 (5 marks)

Nitric oxide is formed during lightning strikes as shown in the reaction below.

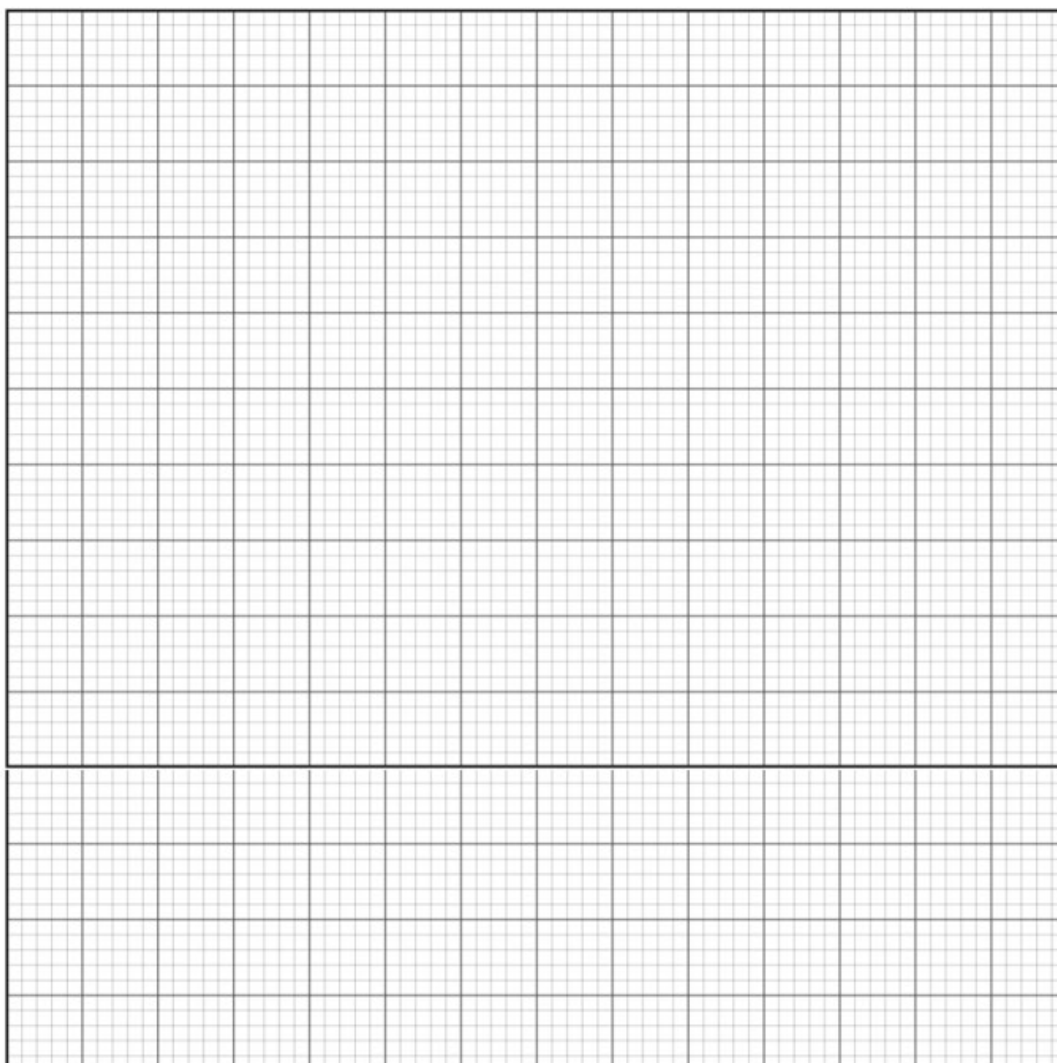


The table below shows the Gibbs Free Energy (ΔG) for this reaction at different temperatures.

Temperature (K)	ΔG (kJ mol ⁻¹)
2000	131
4000	81
6000	31
8000	-19
10000	-69

a) Graph the provided data on the grid below.

3



Question 26 continues on page 21

Question 26 (continued)

(b) Use the graph to estimate the minimum temperature in $^{\circ}\text{C}$ at which this reaction will become spontaneous. Justify your answer. **2**

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End of Question 26

Question 27 (6 marks)

Initially, a mixture has 0.400 mol each of CO_2 and H_2 in a 2.00 L vessel kept at 980°C .

The reaction is $\text{CO}_2(g) + \text{H}_2(g) \rightleftharpoons \text{CO}(g) + \text{H}_2\text{O}(g)$ $\Delta H = +41.2 \text{ kJ mol}^{-1}$

and at this temperature the equilibrium concentration of CO is 0.110 mol L^{-1} .

(a) Write the equilibrium constant expression for this reaction. **1**

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(b) What is the equilibrium concentration of H_2 ? **2**

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(c) Calculate the equilibrium constant at 980°C . **1**

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(d) How would increasing the temperature affect the equilibrium constant? Explain your answer. **2**

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Question 28 (7 marks)

Lead (II) carbonate is a sparingly soluble salt.

- (a) Calculate the molar solubility of lead (II) carbonate in water at 25°C.
Show all working.

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- (b) Quantitatively compare the solubility of lead (II) carbonate in water with its solubility in a 0.10 mol/L solution of lead (II) nitrate, at 25°C.

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- (c) 25 mL of 0.015 M lead (II) nitrate solution and 50 mL of 0.0030 M sodium carbonate solution are mixed. Does precipitation of lead (II) carbonate occur?

Support your answer with calculations

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Question 29 (7 marks)

The solubility of highly soluble, thermally unstable salts such as ammonium chloride may be determined by back titration.

In one experiment a 5.00 mL saturated solution of ammonium chloride, NH_4Cl , at $20.0\text{ }^\circ\text{C}$, was diluted with distilled water to 250.0 mL in a volumetric flask.

A 20.0 mL aliquot of this solution was added to 10.0 mL of 0.400 M NaOH solution. The solution was heated to drive off the ammonia formed by this reaction.

When the flask had cooled, the excess hydroxide ions were neutralised by 14.7 mL of 0.125 M HCl solution. The molar mass of ammonium chloride is 53.5 g mol^{-1} .

(a) Write an equation for the neutralisation reaction. **1**

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(b) Calculate the solubility, in g L^{-1} , of ammonium chloride in water at $20\text{ }^\circ\text{C}$. Show all relevant working. **4**

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(c) If the burette was rinsed with water instead of acid before the titration, how would the calculated solubility of ammonium chloride be affected?
Explain your answer. **2**

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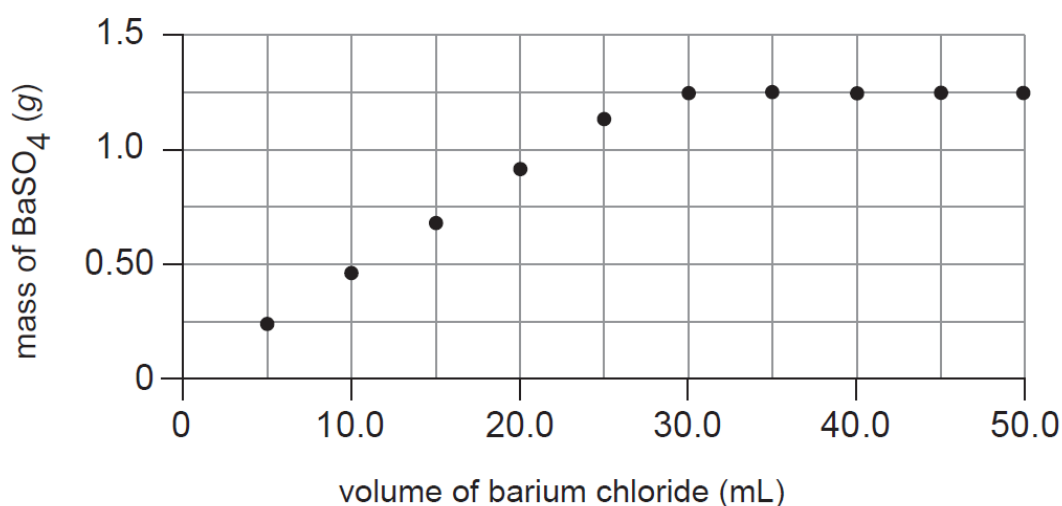
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Question 30 (4 marks)

A gravimetric analysis was undertaken to determine the percentage, by mass, of sulfur in a sample of lawn fertiliser. A 10.4 g sample of fertiliser containing sulfate was dissolved into 250.0 mL of distilled water. This solution was then divided into ten equal 25.0 mL sub-samples.

Various volumes of barium chloride solution was added to each 25.0 mL sub-sample, and the resulting precipitate of barium sulfate was collected by filtration, dried and weighed. The graph shows the results of the gravimetric analysis.



Determine the percentage of sulfur, by mass, in the original sample of lawn fertiliser. Show all relevant working.

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Question 31 (5 marks)

A student was given the organic compounds shown in the table below.

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<i>Compound</i>	<i>Molar mass (g mol^{-1})</i>
but-1-ene	56
1-fluoropropane	62
propan-1-ol	60
ethanamide	59

Predict and explain the different water solubilities of these compounds. Support your answer with at least ONE labelled diagram using structural formulas.

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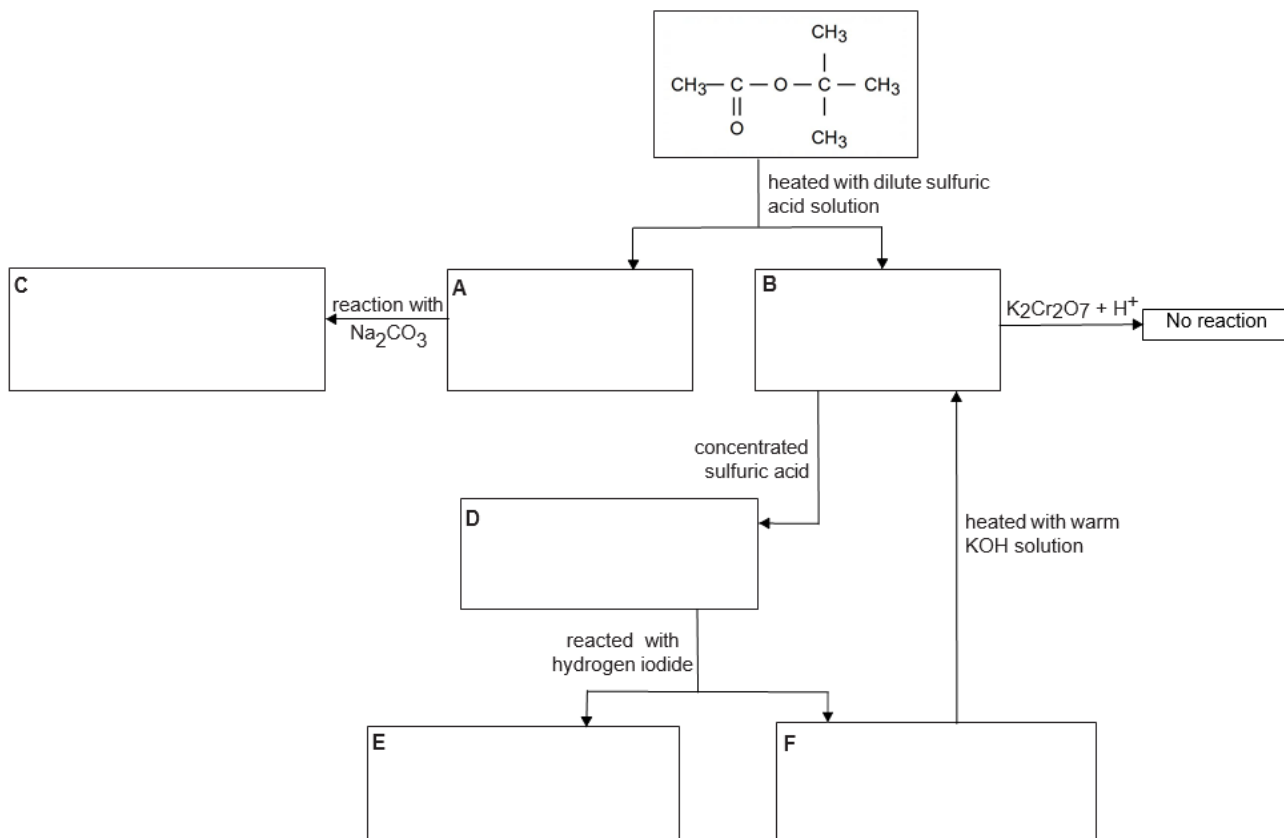
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Question 32 (6 marks)

Esterification is a reversible reaction that can undergo hydrolysis when heated with dilute sulfuric acid solution to drive the reaction towards the reactants.

The flow chart shows reactions involving organic compounds, A to F



Draw the structure of each compound, A to F, in the corresponding space provided.

6

A	B
C	D
E	F

Question 33 (10 marks)

The table shows acid dissociation constants at 25°C

Acid	Formula	K_a
Nitrous	HNO_2	7.2×10^{-4}
Hydrofluoric	HF	6.6×10^{-4}
Hydrocyanic	HCN	6.2×10^{-10}
Propanoic	$\text{CH}_3\text{CH}_2\text{COOH}$	1.34×10^{-5}
Chlorous	HClO_2	1.1×10^{-2}

(a) Identify the weakest acid in the table and determine the $\text{p}K_a$ value for this acid at 25°C **2**

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(b) Identify the weakest base formed from these acids. What is the $\text{p}K_b$ for this base at 25°C? **2**

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(c) Calculate the pH of a 0.010 M solution of propanoic acid at 25°C **2**

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Question 33 continues on page 29

Question 33 (continued)

(d) A buffer solution is prepared by combining 100 mL of 0.010 M nitrous acid and 80 mL of 0.010 M NaNO_2 .

Calculate the pH of this buffer solution and explain the classification of the solution as a buffer.

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End of Question 33

Question 34 (9 marks)

A series of chemical reactions can be used to distinguish between a variety of organic liquids.

(a) Draw a flowchart which could be used to distinguish between the following organic liquids. Include all reagents used and observations made.

4

- HEX-1-ENE
- HEXANE
- PROPAN-1-OL
- PROPAN-2-OL

Question 34 continues on page 31

Question 34 (continued)

(b) Write balanced equations for TWO reactions used in your flowchart using structural formulae.

2

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(c) Predict which of the 4 organic liquids would have the highest boiling point. Explain your prediction.

3

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End of paper

Section II extra writing space

If you use this space, clearly indicate which questions you are answering.

[illegible]

Section II extra writing space

If you use this space, clearly indicate which questions you are answering.

[illegible]

Section II extra writing space

If you use this space, clearly indicate which questions you are answering.

[illegible]

Student Number:

Chemistry Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question.

Fill in the response space completely. If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word **correct** and drawing an arrow.

Start Here	1.	A ☐	B ☐	C ☐	D ☐
	2.	A ☐	B ☐	C ☐	D ☐
	3.	A ☐	B ☐	C ☐	D ☐
	4.	A ☐	B ☐	C ☐	D ☐
	5.	A ☐	B ☐	C ☐	D ☐
	6.	A ☐	B ☐	C ☐	D ☐
	7.	A ☐	B ☐	C ☐	D ☐
	8.	A ☐	B ☐	C ☐	D ☐
	9.	A ☐	B ☐	C ☐	D ☐
	10.	A ☐	B ☐	C ☐	D ☐
	11.	A ☐	B ☐	C ☐	D ☐
	12.	A ☐	B ☐	C ☐	D ☐
	13.	A ☐	B ☐	C ☐	D ☐
	14.	A ☐	B ☐	C ☐	D ☐
	15.	A ☐	B ☐	C ☐	D ☐
	16.	A ☐	B ☐	C ☐	D ☐
	17.	A ☐	B ☐	C ☐	D ☐
	18.	A ☐	B ☐	C ☐	D ☐
	19.	A ☐	B ☐	C ☐	D ☐
	20.	A ☐	B ☐	C ☐	D ☐

2022 CHEMISTRY TRIAL MARKING GUIDELINES

MULTIPLE CHOICE

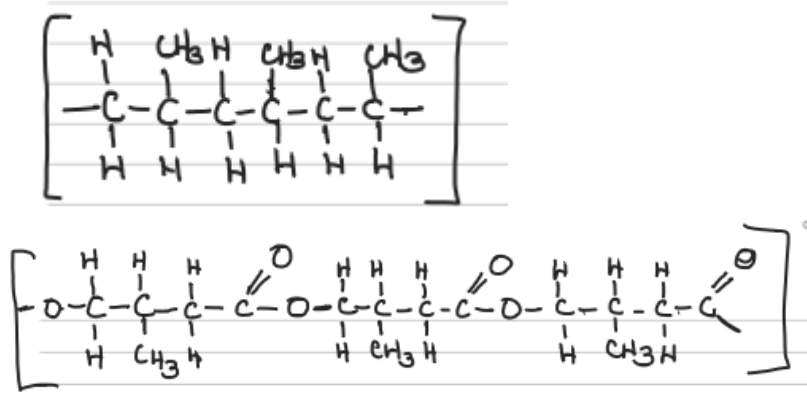
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
B	B	D	C	B	B	B	D	D	B	A	C	D	A	C	A	C	C	C	A

Q15 RE-mark

	Marking Guideline	Suggested Answer							
21 a)	<table><tr><th>Criteria</th><th>Marks</th></tr><tr><td> - Correct direction from 0,0 AND Correct ratio at equilibrium </td><td>2</td></tr><tr><td> - Correct direction from 0,0 OR Correct ratio at equilibrium </td><td>1</td></tr></table>	Criteria	Marks	Correct direction from 0,0 AND Correct ratio at equilibrium	2	Correct direction from 0,0 OR Correct ratio at equilibrium	1		
Criteria	Marks								
Correct direction from 0,0 AND Correct ratio at equilibrium	2								
Correct direction from 0,0 OR Correct ratio at equilibrium	1								
21 b)	<table><tr><th>Criteria</th><th>Marks</th></tr><tr><td> - Correct slope from 0,0 AND Correct ratio at equilibrium </td><td>2</td></tr><tr><td> - Correct slope from 0,0 OR Correct ratio at equilibrium </td><td>1</td></tr></table>	Criteria	Marks	Correct slope from 0,0 AND Correct ratio at equilibrium	2	Correct slope from 0,0 OR Correct ratio at equilibrium	1		CO needed to clearly have the a faster reaction
Criteria	Marks								
Correct slope from 0,0 AND Correct ratio at equilibrium	2								
Correct slope from 0,0 OR Correct ratio at equilibrium	1								
22 a)	<table><tr><th>Criteria</th><th>Marks</th></tr><tr><td> Identifies a relevant risk from the diagram AND Suggests a suitable piece of alternative equipment </td><td>2</td></tr></table>	Criteria	Marks	Identifies a relevant risk from the diagram AND Suggests a suitable piece of alternative equipment	2	- Use of Bunsen burner with flammable liquids - Use hotplate	Most significant hazard Could not get full marks if flammable liquids and open flame was not identified. Only one mark allowed for any other risk correctly identified and controlled.		
Criteria	Marks								
Identifies a relevant risk from the diagram AND Suggests a suitable piece of alternative equipment	2								

	- Identifies a relevant risk from the diagram OR - Suggests a suitable piece of alternative equipment	1		
22 b)	Criteria	Marks	The water cooled condenser reduces loss of reactants and products to the atmosphere by condensing any vapourised components so that they return to reaction vessel as liquid.	Reflux was not accepted as not 'outline'
	Correctly Outlines how 'X' maximises yield	1		
22 c)	Criteria	Marks	One of the following: - Add a small amount of concentrated H₂SO₄ will act as a dehydrating agent and drive reaction to the products - Add excess of one of the reactants to drive equilibrium to the right - Heat reactants on hot plate/heating mantle for an extended time	Ror = rate of reaction Rate of reaction reasons are not suitable as they do not related to yield but rate of yield. Needed to relate to LCP to increase forward reaction Marble chips not accepted as usually for a safety issue. Marble chips should not be used with conc. H ₂ SO ₄ as stops the chips from working and causes a different reaction Must be concentration h ₂ so ₄
	Identifies an appropriate strategy for increasing yield	1		
23 a)	Criteria:	Marks	$\text{C}_6\text{H}_{12}\text{O}_6(s) \xrightarrow{\text{yeast}} 2 \text{C}_2\text{H}_5\text{OH}(l) + 2 \text{CO}_2(g)$	State symbols not marked Zymase accepted
	Correct equation	1		

23 b)	Criteria:	Marks	Any 2 of the following Presence of yeast Absence of oxygen-anaerobic Temperature 37 °C (allow a range) Aqueous solutions	'warm conditions' not accepted as not specific enough Sunlight was not accepted
	• 2 reasonable conditions	2		
	• 1 condition	1		
23 c)	Criteria:	Marks	MM C ₆ H ₁₂ O ₆ = 180 1 mole glucose produced 2 moles ethanol 1250/180 = 6.6667 moles glucose 2x 6.6667 = 13.3333 moles ethanol 13.3333 x 46 = 613.33 g ethanol	Error carried forward from balanced equation MM C ₆ H ₁₂ O ₆ = 180.156 No rounding on MM for full marks 1 mole glucose produced 2 moles ethanol 1250/180.156 = 6.9384311 moles glucose 2x 6.9384311 = 13.87686 moles ethanol 13.87686 x 46.068 = 639.28g ethanol
	• Correct mass value and correct working shown	2		
	• One correct step	1		
24	Criteria:	Marks	A substance is amphiprotic if it can, in the right circumstances, lose a proton (hydrogen ion) to act as a Brønsted–Lowry acid and gain a proton to act as a Brønsted–Lowry base. An example of an amphiprotic substance is sodium hydrogen carbonate, commonly known as bicarbonate of soda. The following reaction occurs when sodium hydrogen carbonate acts as an acid. $\text{HCO}_3^-(aq) + \text{OH}^-(aq) \leftrightarrow \text{CO}_3^{2-}(aq) + \text{H}_2\text{O}(l)$ The following reaction occurs when sodium hydrogen carbonate acts as a base. $\text{HCO}_3^-(aq) + \text{H}_3\text{O}^+(aq) \leftrightarrow \text{H}_2\text{CO}_3(aq) + \text{H}_2\text{O}(l)$	If students used two amphiprotic species they could not get full marks –
	• Uses a suitable example	3		
	• Provides suitable equations.			
	• Gives a detailed explanation			
	• Uses a suitable example	2		
	• Provides at least ONE suitable equation.			
	• Gives an explanation			
• Provides some relevant information	1			

25 a)	Criteria:	Marks	Renewable biodegradable		Important that you read the question, it asked what the advantage of using PHB was as a replacement for PP. Generally, well answered
	Outlines one appropriate advantage	1			
25 b)	Criteria:	Marks	Polymer	Type of polymerisation	This was answered well across the cohort
	Correctly identifies both polymer types	2	PHB	condensation	
	Correctly identifies 1 polymer type	1	PP	addition	
25 c)	Criteria:	Marks			Very poorly answered many students got 0 for this question. PP was the structure that most struggled with PHB many omitted O where the monomers joined. Others included the OH at both ends. You needed to draw 3 units within the polymer as shown in the diagram. If you drew the polymers correctly for both but did not indicate that they would continue, then 1 mark was deducted.
	Correctly draws both polymers with 3 repeating units	2			
	- Correctly draws at least 1 polymer with three repeating units OR - Draws 2 polymers with minor error	1			

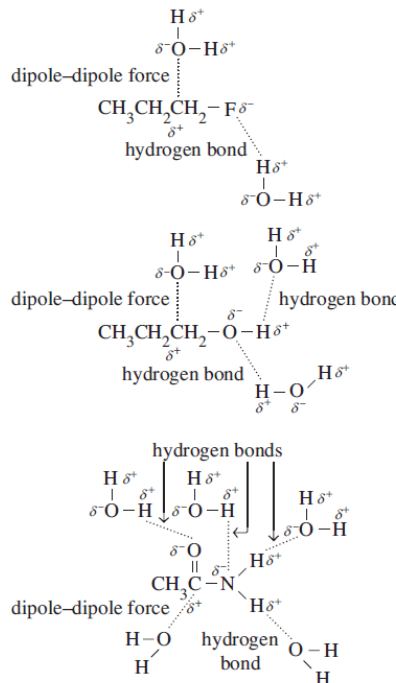
26 a)	<table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td> - Labels axes correctly including units - Uses appropriate scale - Plots points correctly - Draws appropriate line of best fit </td><td>3</td></tr><tr><td> - Provides a substantially correct graph </td><td>2</td></tr><tr><td> - Includes one correct feature </td><td>1</td></tr></table>	Criteria:	Marks	- Labels axes correctly including units - Uses appropriate scale - Plots points correctly - Draws appropriate line of best fit	3	Provides a substantially correct graph	2	Includes one correct feature	1	ΔG versus Temp for $N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$	Generally, very well answered. Common errors here, not labelling axis or excluding units in the axis, incorrect plot points and no line of best fit. Better to use X as a plot point rather than ●, no penalty for this
Criteria:	Marks										
- Labels axes correctly including units - Uses appropriate scale - Plots points correctly - Draws appropriate line of best fit	3										
Provides a substantially correct graph	2										
Includes one correct feature	1										
26 b)	<table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td> - Correct estimate with correct units AND valid justification </td><td>2</td></tr><tr><td> - One correct statement </td><td>1</td></tr></table>	Criteria:	Marks	Correct estimate with correct units AND valid justification	2	One correct statement	1	Temp (°C) $\approx 7200 - 273 = 6927$ This is the temp at which $\Delta G < 0$, and as temp increases from from 6927 °C ΔG will become more negative which indicates a spontaneous reaction.	Very poorly answered, it is important to read the question. You were asked to report in °C, most if the cohort left it in K. Also, you needed to justify that the temperature for spontaneity was greater than 7200 K as when G is negative the reaction becomes spontaneous. If you wrote ≤ 0 this was not awarded as when $G=0$ the reaction is at equilibrium		
Criteria:	Marks										
Correct estimate with correct units AND valid justification	2										
One correct statement	1										
27 a)	<table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td> Correct equilibrium expression </td><td>1</td></tr></table>	Criteria:	Marks	Correct equilibrium expression	1	$K_{eq} = \frac{[CO][H_2O]}{[CO_2][H_2]}$	Common error here was not including the water. All products and reactants are gaseous ECF was applied to subsequent questions if the expression was incorrect				
Criteria:	Marks										
Correct equilibrium expression	1										

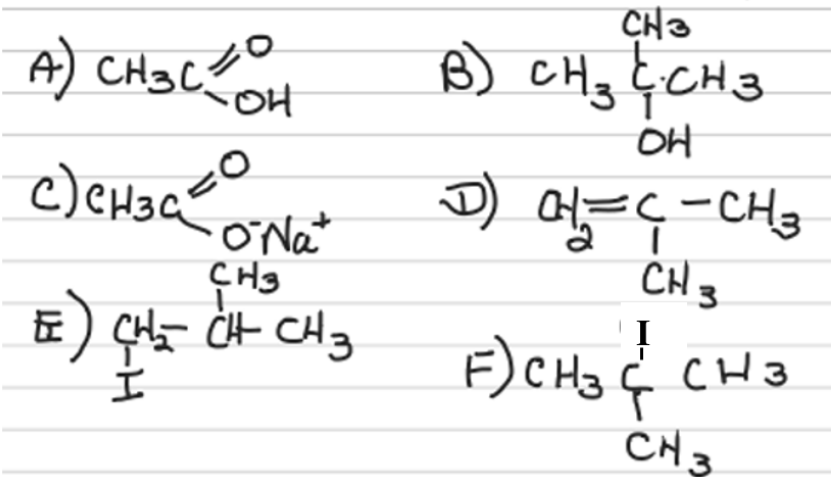
27 b)	Criteria:	Marks	Initial $[\text{CO}_2] = 0.400/2 = 0.200 \text{ mol/L}$ Initial $[\text{H}_2] = 0.400/2 = 0.200 \text{ mol/L}$ <table><tr><td></td><td>CO_2</td><td>H_2</td><td>CO</td><td>H_2O</td></tr><tr><td>I</td><td>0.200</td><td>0.200</td><td>0</td><td>0</td></tr><tr><td>C</td><td>-0.110</td><td>-0.110</td><td>+0.110</td><td>+0.110</td></tr><tr><td>E</td><td>0.090</td><td>0.090</td><td>0.110</td><td>0.110</td></tr></table> Equilibrium $[\text{H}_2] = 0.090 \text{ mol/L}$		CO_2	H_2	CO	H_2O	I	0.200	0.200	0	0	C	-0.110	-0.110	+0.110	+0.110	E	0.090	0.090	0.110	0.110	Common error here was not converting the moles to concentration. Or leaving the moles and using concentration for CO. It is really important that you are either working in moles then, adjusting for volume ie concentration at the end. Alternatively convert to concentration in the initial step. Many lost marks for this reason
		CO_2		H_2	CO	H_2O																		
	I	0.200		0.200	0	0																		
C	-0.110	-0.110	+0.110	+0.110																				
E	0.090	0.090	0.110	0.110																				
- Correct concentration with units for H_2 AND - Calculation/working included in answer	2																							
- Correct concentration with units for H_2 OR - Partial Calculation/working included in answer	1																							
27 (c)	Criteria:	Marks	$\text{Keq} = (0.110 \times 0.110)/(0.090 \times 0.090)$ $= 1.4938$ Or 1.49	Again ECF applied here from the previous question as long as the process was correct																				
	Correct value for Keq (error carried forward from (b) or (c))	1																						
27 (d)	Criteria:	Marks	Forward reaction is endothermic so concentration of products CO and H_2O would increase as equilibrium shifts to the right when temperature increases. Therefore, Keq for the forward reaction would increase when temperature increases.	Well answered. Again, you needed to answer the question																				
	- Correct effect on Keq stated AND - Valid reason for affect	2																						
	One correct statement	1																						
28 (a)	Criteria:	Marks	Let the solubility of PbCO_3 be s. $\text{Ksp PbCO}_3 = 7.40 \times 10^{-14} = [\text{Pb}^{2+}] [\text{CO}_3^{2-}] = (s)^2$ $s^2 = 7.40 \times 10^{-14}$ $s = 2.72 \times 10^{-7} \text{ mol/L}$ The solubility of PbCO_3 in water = $2.72 \times 10^{-7} \text{ mol/L}$	Some students used the incorrect Ksp value and others the incorrect formula for PbCO_3 I even saw a few use the Kw value. Careful when reading data from the data sheet, highlight or underlining may avoid transcription errors																				
	Correct answer, with correct units, showing correct working/reasoning	2																						

	Correct method and reasoning, with mathematical error	1									
28 (b)	<table><tr><td>Criteria:</td><td>Marks</td></tr><tr><td> - Correct conclusion based on correct calculations </td><td>2</td></tr><tr><td> - Correct calculation for the solubility of PbCO₃ in lead (II) nitrate solution </td><td>1</td></tr></table>	Criteria:	Marks	Correct conclusion based on correct calculations	2	Correct calculation for the solubility of PbCO₃ in lead (II) nitrate solution	1	Let the solubility of PbCO₃ in a 0.10 mol/L solution of lead (II) nitrate = x K_{sp} PbCO₃ = [Pb²⁺] [CO₃²⁻] = (x + 0.10) (x) = 7.40 x 10⁻¹⁴ Since x is small by comparison with 0.10 K_{sp} PbCO₃ = [Pb²⁺] [CO₃²⁻] = (0.10) (x) = 7.40 x 10⁻¹⁴ x = 7.40 x 10⁻¹⁴/ (0.10) = 7.4 x 10⁻¹³ mol/L The solubility of PbCO₃ in 0.10 M Pb²⁺ = 7.4 x 10⁻¹³ mol/L The solubility of PbCO₃ in water = 2.72 x 10⁻⁷ mol/L (from part (a) above) Hence PbCO₃ is less soluble in a 0.10 mol/L solution of Pb²⁺ ion than in water.	Overall, well answered. You did need to answer both parts of the question and compare the solubilities. This was a common ion question. A significant portion of the cohort struggled.		
Criteria:	Marks										
Correct conclusion based on correct calculations	2										
Correct calculation for the solubility of PbCO₃ in lead (II) nitrate solution	1										
28 (c)	<table><tr><td>Criteria:</td><td>Marks</td></tr><tr><td> - Correct conclusion based on correct calculations </td><td>3</td></tr><tr><td> - Correct calculation for the [Pb²⁺] [CO₃²⁻] after mixing </td><td>2</td></tr><tr><td> - Correct calculation of no. of moles of Pb(NO₃)₂ and Na₂CO₃ </td><td>1</td></tr></table>	Criteria:	Marks	Correct conclusion based on correct calculations	3	Correct calculation for the [Pb²⁺] [CO₃²⁻] after mixing	2	Correct calculation of no. of moles of Pb(NO₃)₂ and Na₂CO₃	1	No. of moles Pb(NO₃)₂ = 0.015 x (25/1000) = 3.75 x 10⁻⁴ no. of moles Na₂CO₃ = 0.0030 x (50/1000) = 1.5 x 10⁻⁴ Total volume of solution = 75 mL [Pb²⁺] = 3.75 x 10⁻⁴ / 0.075 = 5.0 x 10⁻³ mol/L [CO₃²⁻] = 1.5 x 10⁻⁴ / 0.075 = 2.0 x 10⁻³ mol/L The product of the concentration of the ions is [Pb²⁺] [CO₃²⁻] = 5.0 x 10⁻³ x 2.0 x 10⁻³ = 1.0 x 10⁻⁵ Since this product is greater than K_{sp} (7.40 x 10⁻¹⁴), a precipitate will form.	The students who found 28b challenging did this well. It was familiar to most students. It was obvious that many had practiced this style of question. Common error here was not calculating the new concentrations. You also needed the correct statement based on the data produced.
Criteria:	Marks										
Correct conclusion based on correct calculations	3										
Correct calculation for the [Pb²⁺] [CO₃²⁻] after mixing	2										
Correct calculation of no. of moles of Pb(NO₃)₂ and Na₂CO₃	1										

29 a)	<table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td> Writes appropriate balanced equation with states </td><td>1</td></tr></table>	Criteria:	Marks	Writes appropriate balanced equation with states	1	$\text{NaOH(aq)} + \text{HCl(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$ OR $\text{OH}^-(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{H}_2\text{O(l)}$ $\text{NH}_4\text{Cl} + \text{NaOH} \rightarrow \text{NH}_3 + \text{NaCl} + \text{H}_2\text{O}$	Most girls, but not all wrote a correct equation. Remember water is a liquid. Some misinterpreted and wrote the equation for the other reaction. Remember to read and highlight the question before answering it. Accepted both as the first reaction is also a neutralisation						
Criteria:	Marks												
Writes appropriate balanced equation with states	1												
29 b)	<table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td> - Performs correct calculations - Reports to 3 significant figures </td><td>4</td></tr><tr><td> - Provides main steps of the calculation </td><td>3</td></tr><tr><td> - Provides some relevant steps of the calculation </td><td>2</td></tr><tr><td> - Provides some relevant information </td><td>1</td></tr></table>	Criteria:	Marks	- Performs correct calculations - Reports to 3 significant figures	4	Provides main steps of the calculation	3	Provides some relevant steps of the calculation	2	Provides some relevant information	1	$n(\text{NaOH}) = 0.400 \times 10.0 \times 10^{-3} = 4.00 \times 10^{-3} \text{ (mol)}$ $n(\text{NaOH})$ in excess = $n(\text{HCl})$ $= 0.125 \times 14.7 \times 10^{-3} \text{ mol}$ $= 1.8375 \times 10^{-3} \text{ mol}$ $n(\text{NH}_4\text{Cl})$ in 20.0 mL aliquot = $4.00 \times 10^{-3} - 1.84 \times 10^{-3}$ $= 2.1625 \times 10^{-3} \text{ mol}$ $n(\text{NH}_4\text{Cl})$ in 5.00 mL saturated solution = $n(\text{NH}_4\text{Cl})$ in 250 mL diluted solution $= (n(\text{NH}_4\text{Cl}) \text{ in } 20 \text{ mL diluted solution}/20) \times 250$ $= (2.1625 \times 10^{-3} \div 20) \times 250$ $= 0.02703125 \text{ mol}$ $n(\text{NH}_4\text{Cl})$ in one litre = $(0.02703125/5) \times 1000$ $= 5.40625 \text{ mol}$ <i>Alternative method</i> $n(\text{NH}_4\text{Cl}) = 2.1625 \times 10^{-3} \text{ mol}$ <i>$c \text{ NH}_4\text{Cl}$ diluted sample = $2.1625 \times 10^{-3} \text{ mol}/0.020 = 0.108125 \text{ mol/L}$</i> <i>$c(\text{NH}_4\text{Cl})$ original sample = $0.108125 \text{ mol/L} \times 250/5(\text{dilution factor}) = 5.40625 \text{ mol/L}$</i> $m(\text{NH}_4\text{Cl})$ in one litre = 5.40625×53.5 $= 289 \text{ g}$ Solubility 289 g L^{-1} report to 3 significant figures	Reasonably well done but most students did not get 4 marks for this question. The dilution factor was often forgotten and there were lots of mathematical errors. Don't forget to highlight the figures you need in a response, so you don't forget them. The dilution factor was a little unusual so watch that you use this properly.
Criteria:	Marks												
- Performs correct calculations - Reports to 3 significant figures	4												
Provides main steps of the calculation	3												
Provides some relevant steps of the calculation	2												
Provides some relevant information	1												
29 c)	<table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td> - Correctly identifies larger titre would result - Appropriate explanation of how the increased titre </td><td>2</td></tr></table>	Criteria:	Marks	- Correctly identifies larger titre would result - Appropriate explanation of how the increased titre	2	Since the HCl(aq) in the burette was diluted, a larger $V(\text{HCl})$ (titre) was required to react with the excess NaOH . Hence the $n(\text{NaOH})$ in excess was calculated as higher than true, leading to a smaller calculated $n(\text{NaOH})$ reacted with NH_4Cl and a lower calculated amount (solubility) of NH_4Cl .	This question required an explanation of higher or lower. The best responses showed how this dilution affected each step.						
Criteria:	Marks												
- Correctly identifies larger titre would result - Appropriate explanation of how the increased titre	2												

	<table><tr><td>volume results in a lower calculated solubility</td><td></td></tr><tr><td> - Identifies larger titre volume AND/OR - Limited explanation </td><td>1</td></tr></table>	volume results in a lower calculated solubility		- Identifies larger titre volume AND/OR - Limited explanation	1								
volume results in a lower calculated solubility													
- Identifies larger titre volume AND/OR - Limited explanation	1												
30	<table><tr><td>Criteria:</td><td>Marks</td></tr><tr><td> - Determines the percentage of sulfur. - Performs correct calculation. - Shows all relevant working </td><td>4</td></tr><tr><td> - Provides main steps of the calculation </td><td>3</td></tr><tr><td> - Provides some relevant steps of the calculation </td><td>2</td></tr><tr><td> - Provides some relevant information </td><td>1</td></tr></table>	Criteria:	Marks	- Determines the percentage of sulfur. - Performs correct calculation. - Shows all relevant working	4	Provides main steps of the calculation	3	Provides some relevant steps of the calculation	2	Provides some relevant information	1	The mass of fertiliser in each 25.0 mL sub-sample is 1.04 g. From the graph, the maximum mass of BaSO₄ is 1.25 g. molar mass of BaSO₄ = 137.3 + 32.07 + 4x16.00 =233.4 g mol⁻¹ $n(\text{BaSO}_4) = 1.25/233.4 = 0.00536 \text{ mol}$ $n \text{ S} = 0.00536 \text{ mol}$ mass of sulfur 0.00536 x 32.07 = 0.172 g % of sulfur = 0.172/1.04 x100 = 16.5%	Many girls did not distinguish between sulfur and sulfate. Many also did not read correctly from the graph. Be careful about what the question is asking.
Criteria:	Marks												
- Determines the percentage of sulfur. - Performs correct calculation. - Shows all relevant working	4												
Provides main steps of the calculation	3												
Provides some relevant steps of the calculation	2												
Provides some relevant information	1												

31	Criteria:	Marks	Water can form hydrogen bonds and dipole–dipole forces with neutral molecules. All four compounds are neutral molecules that, being similar in size, form similarly sized dispersion forces. • But-1-ene is non-polar and cannot form hydrogen bonds with water. Therefore, but-1-ene is insoluble in water. • 1-fluoropropane is very polar and can form both one hydrogen bond with water. Therefore, it is slightly soluble in water. • Both propan-1-ol and ethanamide are very polar and can form hydrogen bonds and some dipole dipole with water. Both can form many more hydrogen bonds with water than 1-fluoropropane, resulting in both compounds being much more water-soluble than 1-fluoropropane.	While lots of students were awarded 4/5 for this question, overall it was reasonably well done. The major mistake was NOT USING THE DATA given. (this was most students). Some comment about the fact that the substances have similar MM so dispersion forces were similar between water and all the substance. Many girls confused solubility with boiling points. It is important to remember in solubility that the force is with the water molecule. Also lots of girls had OH as an hydroxide (neg) as opposed to an hydroxyl group. Accurate drawings with labels are essential. NB. They were not asked to rank solubility. Also note the correct definition of a hydrogen bond.
	- Explains the different water solubilities of all FOUR compounds. AND - Provides at least ONE appropriate, fully labelled diagram - Refer to DATA	5		
	- Explains the different water solubilities of at least TWO compounds. AND - Provides at least ONE diagram.	3-4		
	- Explains the different water solubilities of at least ONE compound. OR - Outlines some intermolecular forces associated with functional groups	2		
	Provides some relevant information	1		

32	Criteria	Marks		A and B involved finding the carboxylic acid and alcohol formed from the ester Very few students were able to correctly draw the structure of C Some students had E and F reversed 1 mark only was awarded for this Error carried forward was accepted if a structure was drawn correctly from an incorrect prior structure
	Identifies 6 compounds	6		
	Identifies 5 compounds	5		
	Identifies 4 compounds	4		
	Identifies 3 compounds	3		
	Identifies 2 compounds	2		
	Identifies 1 compound	1		
33 a)	Criteria:	Marks	hydrocyanic acid $pK_a = -\log_{10}(K_a) = -\log_{10}(6.2 \times 10^{-10}) = 9.21$ (2 s.f.)	Well done across the cohort
	Identifies hydrocyanic acid as the weakest acid AND determines the pK_a value for hydrocyanic acid	2		
	Identifies hydrocyanic acid as the weakest acid OR determines the pK_a value for the identified acid	1		
33 b)	Criteria:	Marks	HClO_2 (chlorous acid) forms the chlorite ion (ClO_2^-) as its conjugate base. The strongest acid forms the weakest base. For a conjugate acid-base pair: $pK_a = -\log(K_a) = -\log(1.1 \times 10^{-2}) = 1.96$ $pK_b = 14 - pK_a = 14.00 - 1.96 = 12.04$	Some students only mentioned Chlorous acid or HClO_2, a reminder to read the question carefully. The weakest base which is ClO_2^- ion should be stated in answer Most students calculated pK_b correctly
	Identifies the ClO_2^- (chlorite ion) as the weakest base AND determines the pK_b for this base	2		
	Identifies the ClO_2^- (chlorite ion) as the weakest base	1		

	OR determines the pK_b for the identified base																
33 c)	<table><tr><td>Criteria:</td><td>Marks</td></tr><tr><td>• Calculates the correct pH</td><td>2</td></tr><tr><td>• Shows some correct working and reasoning</td><td>1</td></tr></table>	Criteria:	Marks	• Calculates the correct pH	2	• Shows some correct working and reasoning	1	$\text{CH}_3\text{CH}_2\text{COOH} (aq) + \text{H}_2\text{O} (l) \rightleftharpoons \text{CH}_3\text{CH}_2\text{COO}^- (aq) + \text{H}_3\text{O}^+ (aq)$ $K_a \text{ CH}_3\text{CH}_2\text{COOH} = \frac{[\text{CH}_3\text{CH}_2\text{COO}^-] [\text{H}_3\text{O}^+]}{[\text{CH}_3\text{CH}_2\text{COOH}]} = 1.34 \times 10^{-5}$ Let x moles of $\text{CH}_3\text{CH}_2\text{COOH}$ ionise, forming x moles of H_3O^+ $K_a \text{ CH}_3\text{CH}_2\text{COOH} = \frac{[x] [x]}{[0.010 - x]} = 1.34 \times 10^{-5}$ Assume x will be small by comparison with 0.010 Hence $[x]^2 = 1.34 \times 10^{-5} \times 0.010 = 1.34 \times 10^{-7}$ $[\text{H}_3\text{O}^+] = \sqrt{1.34 \times 10^{-7}} = 0.000366 \text{ mol/L}$ Hence $\text{pH} = -\log_{10} (0.000366) = 3.44 \text{ (2 s.f.)}$	Most students answered this question well. Weaker answers calculated pH using the acid concentration given without considering K_a to determine how much acid dissociated/ionised to $[\text{H}_3\text{O}^+]$								
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33 d)	<table><tr><td>Criteria:</td><td>Marks</td></tr><tr><td>• Calculates concentrations of $[\text{NO}_2^-]$ and $[\text{HNO}_2]$</td><td>4</td></tr><tr><td>• Uses equilibrium expression to calculate H_3O^+</td><td></td></tr><tr><td>• Calculates the pH</td><td></td></tr><tr><td>• Explains why this mixture is classified as a buffer (2 points)</td><td></td></tr><tr><td>• 3 of the above</td><td>3</td></tr><tr><td>• 2 of the above</td><td>2</td></tr></table>	Criteria:	Marks	• Calculates concentrations of $[\text{NO}_2^-]$ and $[\text{HNO}_2]$	4	• Uses equilibrium expression to calculate H_3O^+		• Calculates the pH		• Explains why this mixture is classified as a buffer (2 points)		• 3 of the above	3	• 2 of the above	2	$\text{HNO}_2 (aq) + \text{H}_2\text{O} (l) \rightleftharpoons \text{NO}_2^- (aq) + \text{H}_3\text{O}^+ (aq)$ $K_a \text{ HNO}_2 = \frac{[\text{NO}_2^-] [\text{H}_3\text{O}^+]}{[\text{HNO}_2]} = 7.2 \times 10^{-4}$ Initial $[\text{HNO}_2] = 0.010 \text{ mol/L}$ Initial $[\text{NO}_2^-] = 0.010 \text{ mol/L}$ On mixing, the volume is increased to 180 mL After mixing, $[\text{HNO}_2] = 0.010 \times 100/180 = 0.00555 \text{ mol/L}$ After mixing, $[\text{NO}_2^-] = 0.010 \times 80/180 \text{ mol/L} = 0.00444 \text{ mol/L}$ Let x mol/L HNO_2 ionise at equilibrium. At equilibrium $[\text{HNO}_2] = (0.00555 - x) \text{ mol/L}$ $[\text{NO}_2^-] = (0.00444 + x) \text{ mol/L}$ $[\text{H}_3\text{O}^+] = x \text{ mol/L}$ $K_a \text{ HNO}_2 = \frac{[\text{NO}_2^-] [\text{H}_3\text{O}^+]}{[\text{HNO}_2]} = 7.2 \times 10^{-4} = \frac{(0.00444 + x) (x)}{(0.00555 - x)}$	This question was poorly answered. Few students gained full mark for this question. Most students calculated pH by using the concentration of acid given which assumes full dissociation/ionisation, not correct. Equilibrium expression was needed to find $[\text{H}_3\text{O}^+]$ using the K_a value provided in data table. Also some students calculated the moles of each and subtracted to find excess – in this case error carried forward was used to award 1 mark for pH calculation using excess HNO_2
Criteria:	Marks																
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	1 of the above	1	$[\text{HNO}_2]$ (0.00555 - x) Assume x is small by comparison with 0.00555 and with 0.00444 mol/L $K_a \text{ HNO}_2 = \frac{[\text{NO}_2^-][\text{H}_3\text{O}^+]}{[\text{HNO}_2]} = 7.2 \times 10^{-4} = \frac{(0.00444)(x)}{(0.00555)}$ $[\text{H}_3\text{O}^+] = x = 7.2 \times 10^{-4} \times 0.00555 / 0.00444 = 0.00090 \text{ mol/L}$ pH = 3.05 A buffer solution is one which will maintain an almost constant pH, even if small quantities of strong acid or base are added to it. As long as there are close to equal moles of equal concentration solutions making up the buffer mixture, and the acid and base are both only moderately strong as acids and bases, the solution will stay at close to the pH value 3.05, as calculated above. By Le Chatelier's Principle, if [x] is small by comparison with the concentrations of the acid and base, if the concentration of H⁺ in the buffer mixture changes slightly, the proportions of HNO₂ and NO₂⁻ will change to keep the pH close to 3.05.	For 1 mark for buffer explanation you needed at least 2 reasons									
34 a)	<table><thead><tr><th>Criteria</th><th>Marks</th></tr></thead><tbody><tr><td>• Draws a flowchart which correctly distinguishes the 4 compounds</td><td>4</td></tr><tr><td>• Draws a flowchart which correctly distinguishes 3 compounds</td><td>3</td></tr><tr><td>• Draws a flowchart which correctly distinguishes 2 compounds</td><td>2</td></tr><tr><td>• Draws a flowchart which correctly distinguishes 1 compound</td><td>1</td></tr></tbody></table>	Criteria	Marks	• Draws a flowchart which correctly distinguishes the 4 compounds	4	• Draws a flowchart which correctly distinguishes 3 compounds	3	• Draws a flowchart which correctly distinguishes 2 compounds	2	• Draws a flowchart which correctly distinguishes 1 compound	1	<pre>graph TD A[Samples of each of 4 liquids] --> B[Add water] B --> C[Soluble.] B --> D[Insoluble.] C --> E["Lucas Test
anhydrous ZnCl2/HCl (conc)"] E --> F[N.R] E --> G["turbid (3-5 mins)"] F --> H[colourless] H --> I[propan-1-ol] G --> J[propan-2-ol] D --> K["Br2(aq)
absent on light"] K --> L[Br2 decol] K --> M[NR] L --> N[hexene] M --> O[hexane] O --> P[Stays brown/orange]</pre>	Was well done. Problems here: Some students included an oxidation test with acidified KMnO₄ for distinguishing between the alcohols and used indicator or Na₂CO₃. Regardless of whether a carboxylic acid was formed or a ketone, there would be acid present. Testing with oxidation as the first step was a little tricky as hex-1-ene will oxidise as well as the alcohols.
Criteria	Marks												
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34 b)	Criteria	Mark	$\text{HCl} + \text{CH}_3\underset{\text{OH}}{\text{CH}}\text{CH}_3 \xrightarrow{\text{ZnCl}_2} \text{CH}_3\underset{\text{Cl}}{\text{CH}}\text{CH}_3 + \text{H}_2\text{O}$ $\text{CH}_2=\text{CHCH}_2\text{CH}_2\text{CH}_2\text{CH}_3 + \text{Br}_2 \longrightarrow \text{CH}_2\text{BrCHBrCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	Most students were able to correctly write the addition reaction of Br₂ to hex-1-ene. Students lost marks here for not using condensed structural or structural formula Including a 'no reaction; for bromine with hexane was not accepted.
	2 correct reactions using structural formula	2		
	1 correct reaction or 2 partially correct	1		
34 c)	Criteria	Mark	· propan-1-ol has the highest boiling point AND · propan-1-ol has the strongest intermolecular forces because hydrogen bonding as well as dispersion forces exist between the molecules. Hydrogen bonds are the strongest IMFs so it takes more energy to disrupt this force, so a higher BP. Propan-2-ol also has hydrogen bonding because of the OH group, but this OH group is not as exposed so the IMF is less and so the BP is less . AND Hexene and hexane are both non polar so the only IMF that occurs between the molecules is dispersion, which is a weak force.	A lot of students were awarded full marks for this question. Some issues here were: • not including all the molecules • not mentioning the types of IMF present for each molecule • not comparing the strength of IMF
	Predicts that propan-1-ol the highest boiling point AND Differentiates between propan-1-ol and propan-2-ol (hydroxyl more exposed for hydrogen bonding) AND identifies the intermolecular forces in the other molecules and compares their strength	3		
	2 of the above	2		
	1 of the above	1		